

Graphs and Networks (GaN) 2026-2027

Lecture 1 of 14
15 September 2026
Timur Friedman

Audio recording

I will be recording my lectures to help myself to improve the course. These are for my professional use only, and I will not be sharing the recordings publicly.

If you do not wish to be recorded when asking a question or making a comment, please wave your hand, and I will pause the recording.

Graphs and Networks (GaN) course description

The course is divided into two parts of 7 weeks each

Course identifier: UM5IN067 - S1-26

Graphs and Networks (GaN) course description

Networks

Like many large societal constructs, the structure of the Internet is undocumented. It is composed of hundreds of thousands of independently-administered domains that do not readily reveal information about their internal structures or how they are interconnected. The first part of the course (Networks) looks at the network measurement techniques that are available to us, what they reveal, and what remains unknown. This sets the context for understanding the nature of the Internet graphs that we are able to generate and study.

Graphs and Networks (GaN) course description

Graphs

The second part (Graphs) focuses on the graph-based analysis of data from network measurements. The structure of the network is analysed, with the aim of understanding how structural characteristics affect the way the network operates. Each course presents a transversal notion on a specific research problem and illustrates it on the case of networks of machines, in particular the topology of the Internet. This part of the course is at the intersection of data mining and graph algorithms.

Lecturers, teaching assistants, administrator

Graphs

LIP6 Complex Networks (Complex) Team

Lionel Tabourier (course co-lead)

Bastien Legay

Networks

LIP6 Networks and Performance Analysis (NPA) Team Dioptra Research Group

Timur Friedman (course co-lead)

Hugo Rimlinger

Administrator

TBD

Schedule of lectures, labs, and exams

Networks		Graphs		Post-lecture period	
1	15 Sept.	8	10 Nov.	study week	
2	22 Sept.	9	17 Nov.	final exam	9 Feb.?
3	29 Sept.	10	23 Nov.		
4	6 Oct.	11	1 Dec.		
5	13 Oct.	12	8 Dec.		
6	20 Oct.	13	15 Dec.		
Toussaint		Noël			
7	3 Nov.	14	5 Jan.	make-up exam	September – date to be determined
Make-up dates, If needed: 12 and 19 Jan.					

Room assignments, Moodle, e-mail

To find the room assignments, which can change unexpectedly, select M2_RES on the CalDav interface and look for UM5IN067-GAN-cours (courses) and UM5IN067-GAN-TME (labs):

<https://cal.ufr-info-p6.jussieu.fr/master/>

The Moodle is where we will post lecture slides and other information relating to the course, including (new this year) grades. Look for “Graphs and Networks - S1-26” in your course list:

<https://moodle-sciences-26.sorbonne-universite.fr/>

To reach any member of the teaching team, you may write to us at prenom.nom@lip6.fr. You are advised to put “GaN” (without quotation marks) at the beginning of the subject line, to be sure to get the attention of your recipient.

Grading

This is a hands-on course! **40%** of your grade will come from your work in the **computer lab**, the other **60%** from the **final exam**, which will cover both lecture material and things that you will have been expected to have learned in the lab.

For the labs, you will receive a grade for the Graphs part, and a grade for the Networks part, each therefore counting for 20% of your overall grade. You will do lab exercises each week: 7 Graphs exercises, 7 Networks exercises. You can choose the 3 Graphs exercises and the 3 Networks exercises on which you wish to be graded.

If you do not receive a passing grade for the course, you may take the make-up exam, the result of which replaces your final exam grade. If you do find yourself in this situation, please pay careful attention to the administrative rules concerning make-up exams! For details on this issue, you are advised to consult with the Masters program directly.

Where graphs and networks meet

- Cover story of Nature, 27 July 2000

Error and attack tolerance of complex networks

Réka Albert, Hawoong Jeong & Albert-László Barabási

*Department of Physics, 225 Nieuwland Science Hall, University of Notre Dame,
Notre Dame, Indiana 46556, USA*

Many complex systems display a surprising degree of tolerance against errors. For example, relatively simple organisms grow, persist and reproduce despite drastic pharmaceutical or environmental interventions, an error tolerance attributed to the robustness of the underlying metabolic network¹. Complex communication networks² display a surprising degree of robustness: although key components regularly malfunction, local failures rarely lead to the loss of the global information-carrying ability of the network. The stability of these and other complex systems is often attributed to the redundant wiring of the functional web defined by the systems' components. Here we demonstrate that error tolerance is not shared by all redundant systems: it is displayed only by a class of inhomogeneously wired networks,



Overview of the Networks part of the course

We will focus on the internet, labs will focus on questions of IP geolocation, giving you experience with a range of measurement tools and methods.

- Refresher on the essentials of the internet (autonomous systems, packet switching), how it lends itself to measurement and what the main obstacles are
- What is “the graph of the internet”?
- What issues have arisen with respect to this graph?
- Autonomous systems and BGP
- IP level, tunnels, local area networks
- Best measurement practices, ethical issues
- Tools such as ping and traceroute, path lengths, multipath routing
- Vantage points
- Metadata (bandwidth, latency, jitter, loss, geolocation, etc.)

Class 1: Introduction to networks, as related to graphs

What defines the internet, how is it administered, how are its technical aspects decided?

What are the principal protocols of the internet?

What might be meant by the 'network topology'? What aspects of the internet can be considered as a graph?

What might be useful about these graphs?

What are the strictly graph aspects, what is the important metadata?

What are our sources of information? Databases? Measurements, both passive and active?

Different views of “the graph of the internet”

- The graph of autonomous systems
- The graph of independent networks
- The PoP graph
- The router-level graph
- The IP address graph

It is difficult to measure the internet

The internet is opaque

- ▶ Paths between hosts are **not exposed** by the network
- ▶ No central API to query: “what route does a packet take?”
- ▶ Paths emerge from **independent decisions** at hundreds of thousands of routers

Why it is important to measure the internet

- ▶ Network failures, security breaches, and performance degradation all manifest as **path changes**
- ▶ Understanding paths is essential for operators, researchers, and regulators

Measurement difficulties arise from the internet's design

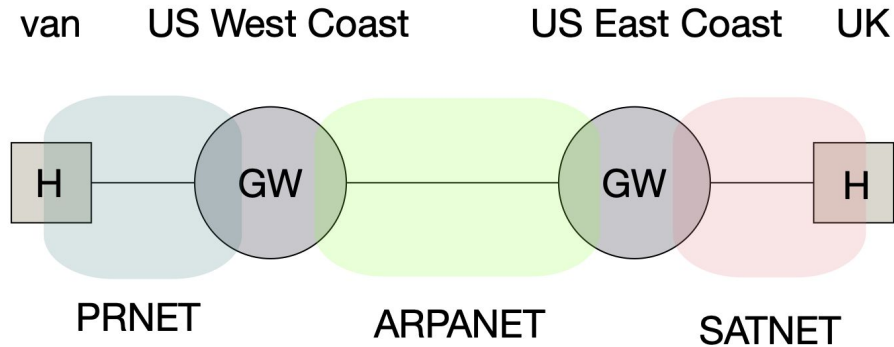
Two key design decisions

- ▶ Packet switching (1970s)
- ▶ Autonomous system organization (1980s)

Advantages and disadvantages

- ▶ Advantage: distributed, resilient design
- ▶ Disadvantage: no central source of information

Cerf & Kahn's 1977 packet switching demo

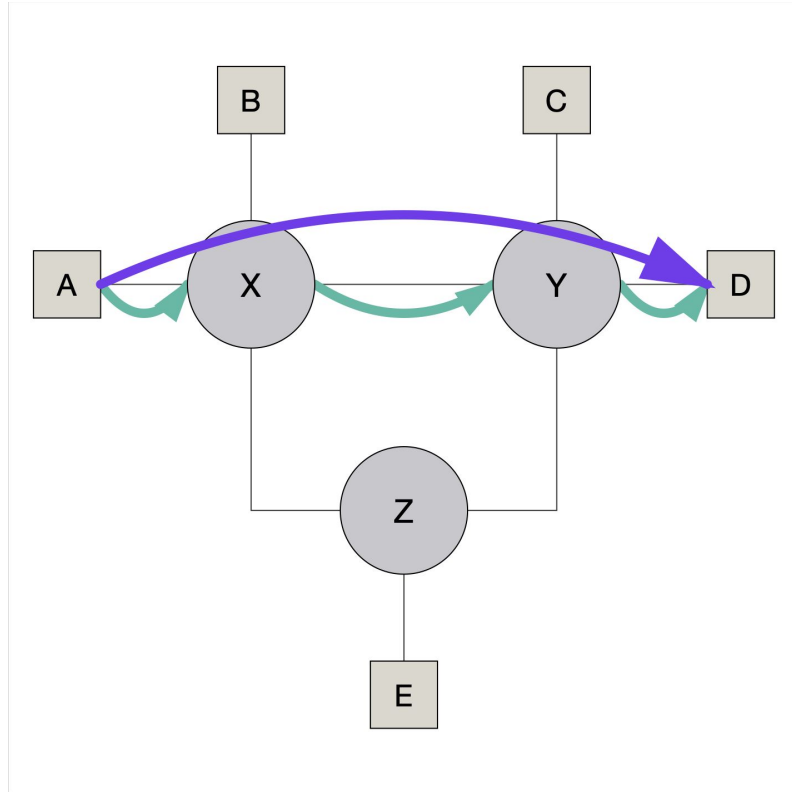


- ▶ Packet switching proof of concept
 - The earliest days of IP (then called TCP!)

- ▶ Heterogeneous networks:
 - PRNET packet radio
 - ARPANET leased lines
 - SATNET satellite radio

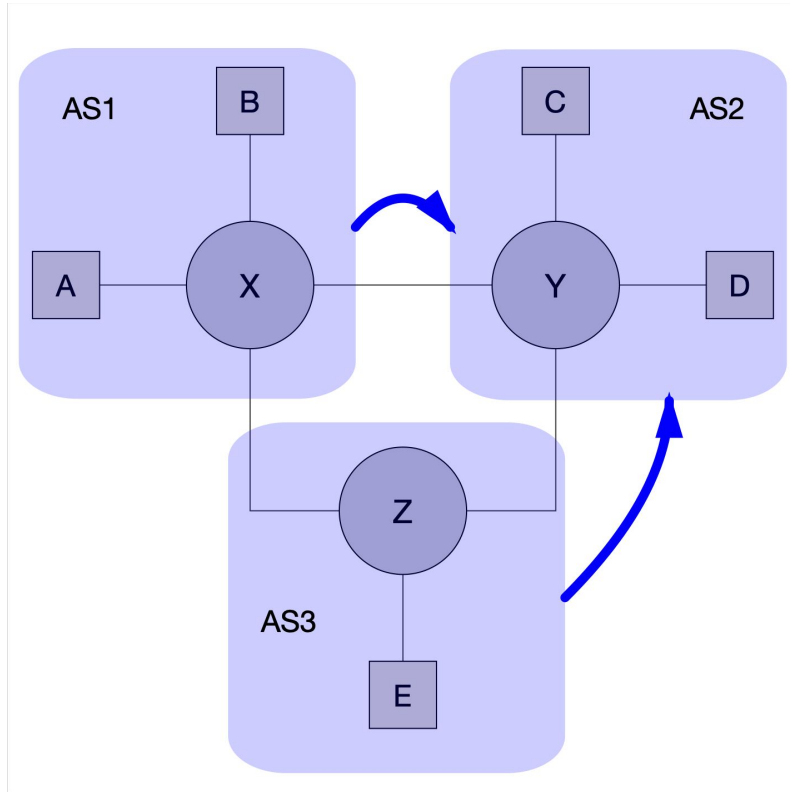
- ▶ Transcontinental and transatlantic
 - host (H) in a van on the US West Coast
 - gateway (GW) within radio range of the van
 - gateway (GW) on the US East Coast
 - host (H) in a lab in the UK

Packet switching: a simpler alternative to circuit switching



- Circuit switching
 - End-to-end circuit: A to X to Y to D
 - X and Y forward based on a circuit ID
- Packet switching
 - A simpler mechanism
 - The end-to-end path arises from a series of local decisions
 - X and Y each forward based on packet destination D
- Measurement difficulty
 - No end-to-end circuit that could be easily queried
 - Each local packet forwarding decision needs to be learned

Autonomous system paths are not all known



- ▶ Autonomous systems (ASes)
 - Introduced in the 1980s
 - Concept: AS path to a destination addresses
- ▶ Border Gateway Protocol (BGP)
 - Global routing algorithm between ASes
 - ASes voluntarily share path information with their neighbors
- ▶ Measurement difficulty
 - Example: AS1 and AS3 might not share their paths with each other
 - No AS learns all of the paths in the internet

Attempting to overcome the measurement difficulties

Packet switching

- ▶ Tracing paths using traceroute
- ▶ “Active measurements”

Autonomous systems

- ▶ Learning paths from route collectors
- ▶ “Passive measurements”

As we shall see, there is a mismatch between the features that we would like to measure and what we are capable of measuring.